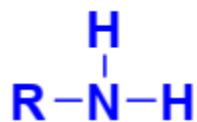


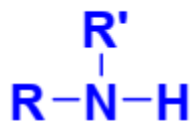


AMINE DIAZONIUM

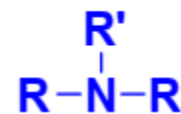
RNH_2



amine bậc 1



amine bậc 2



amine bậc 3

²



Danh pháp

Tên
thông
thường

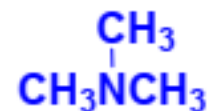
Tên gốc hydrocarbon + amine



methylamine



methylpropylamine



trimethylamine

Tên
IUPAC

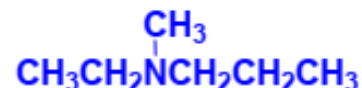
Amine không chứa $-\text{COOH}$, $-\text{CHO}$, $-\text{OH}$: *alkanamine*



butanamine



N-ethyl-3-hexanamine



N-ethyl-*N*-methyl-1-propanamine

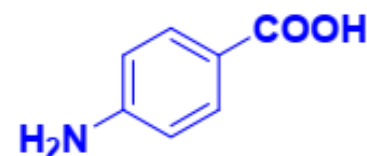
Trật tự ưu tiên: $-\text{COOH} > -\text{CHO} > >\text{C}=\text{O} > -\text{OH} > -\text{NH}_2$



2-aminoethanol



4-amino-2-butanone

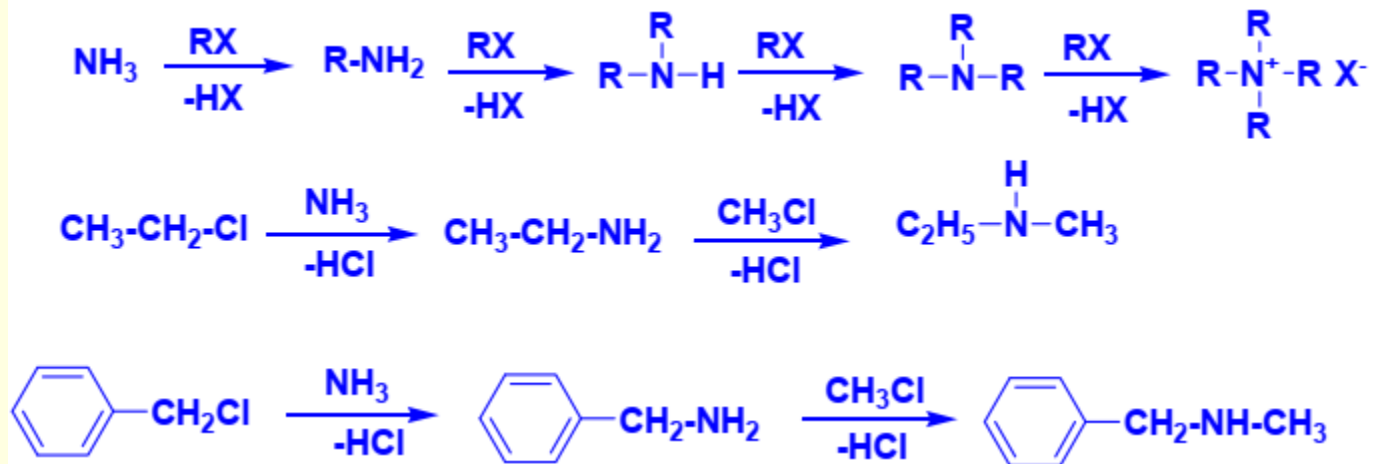


4-aminobenzoic acid

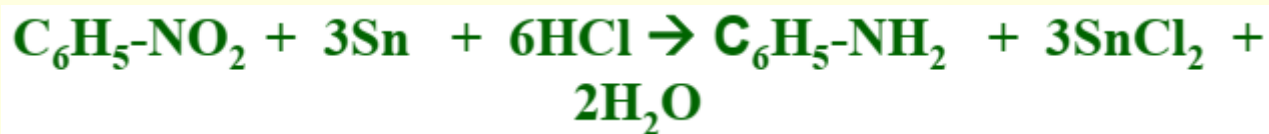
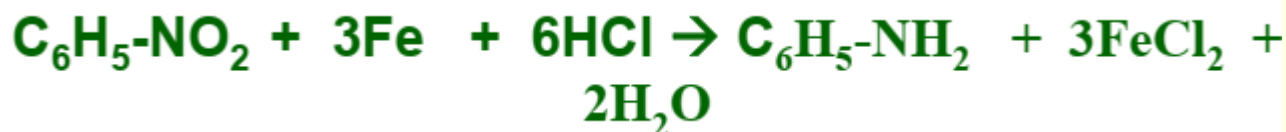
Điều chế



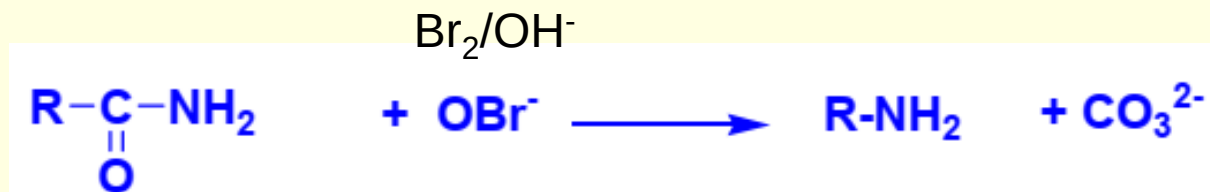
- Alkyl hóa amine



- Khử bằng H mới sinh



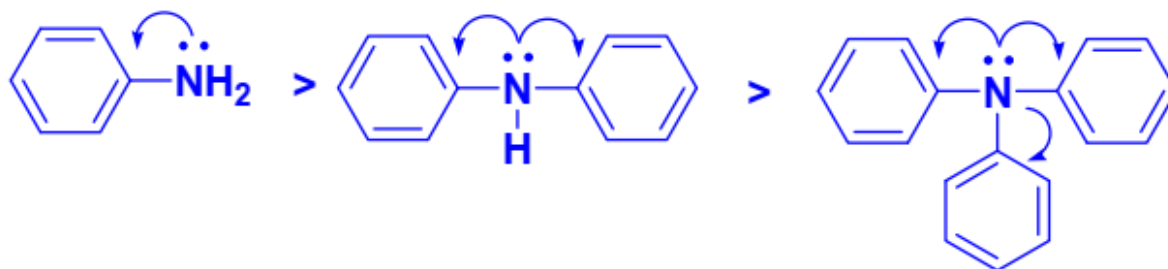
- Chuyển vị Hoffman



Tính base Đẩy → tăng



- **Amine thơm: bậc càng cao, tính base càng giảm**
(do +C của N)

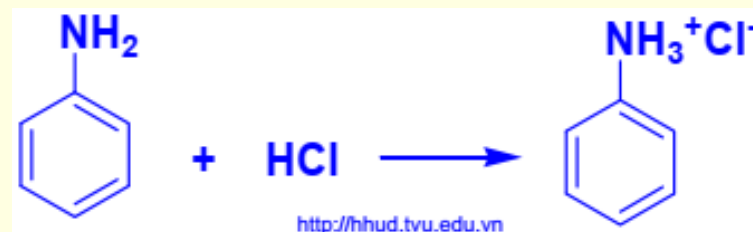


- **Nhóm thế trong nhân thơm đẩy điện tử → tính base tăng và ngược lại**

Tính base: $p\text{-NO}_2\text{-C}_6\text{H}_4\text{-NH}_2 < m\text{-NO}_2\text{-C}_6\text{H}_4\text{-NH}_2$

$< p\text{-Cl-C}_6\text{H}_4\text{-NH}_2 < \text{C}_6\text{H}_5\text{-NH}_2$

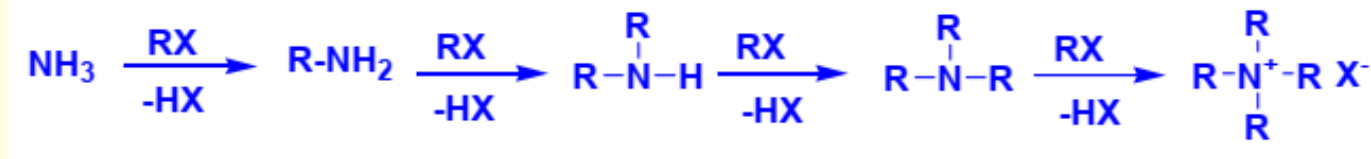
$< p\text{-CH}_3\text{O-C}_6\text{H}_4\text{-NH}_2$



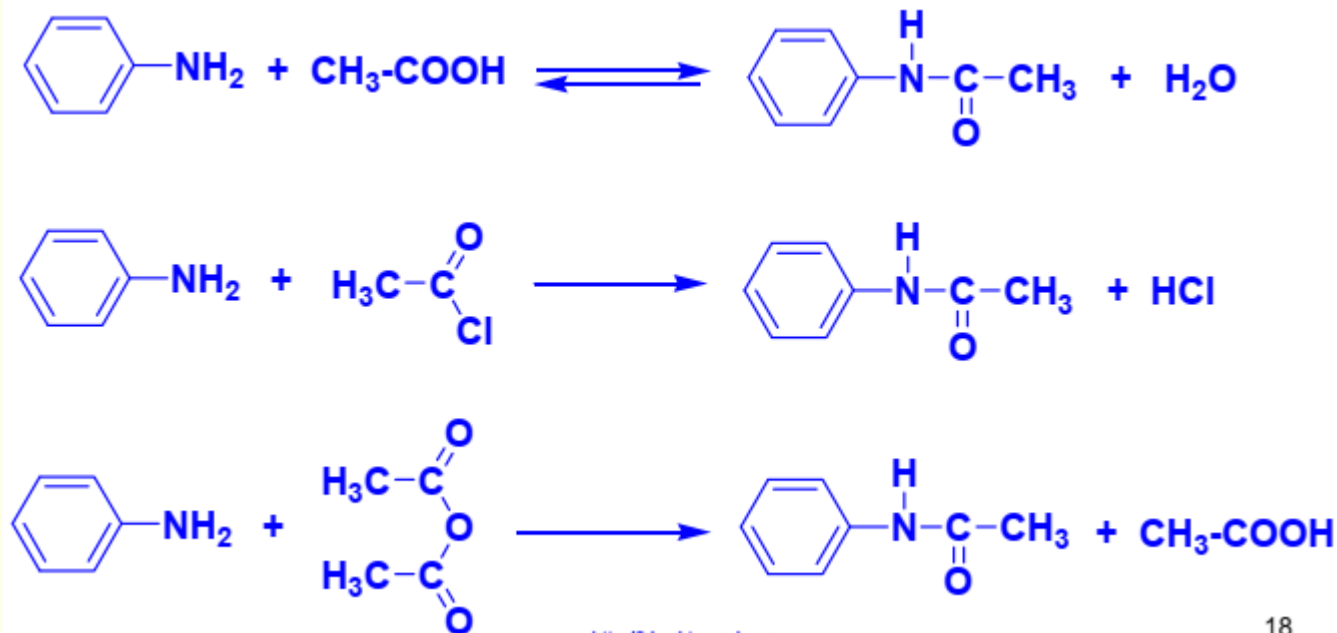
Tính chất hóa học



■ Alkyl hóa



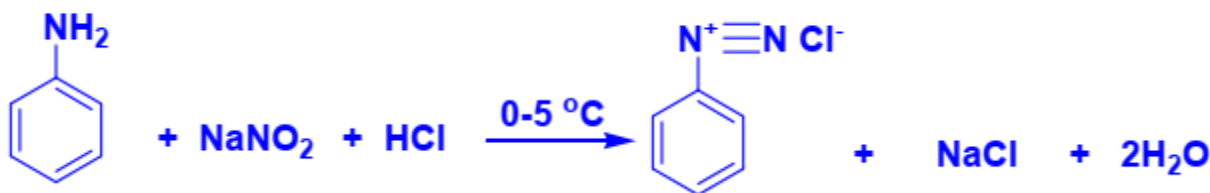
■ Acyl hóa



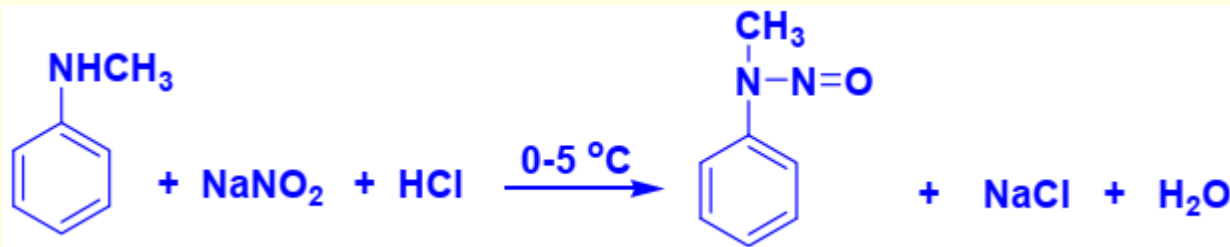
Tính chất hóa học



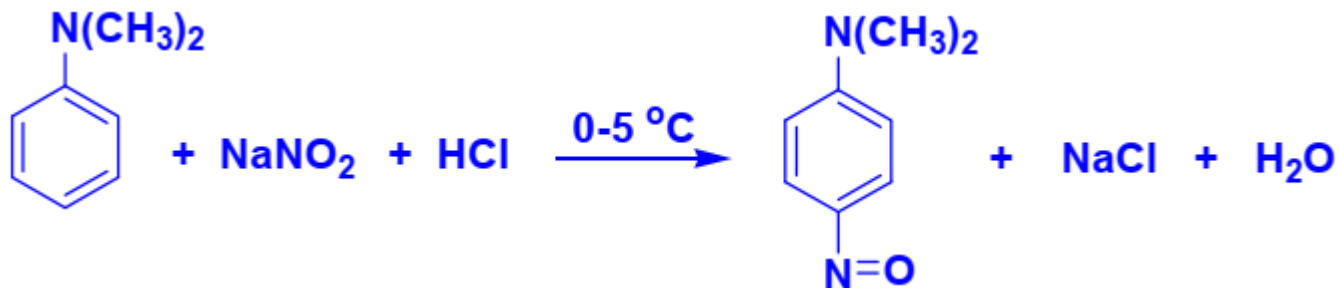
- Tạo muối diazonium (amine thơm bậc 1)



- Amine bậc 2
→ nitrosoamine

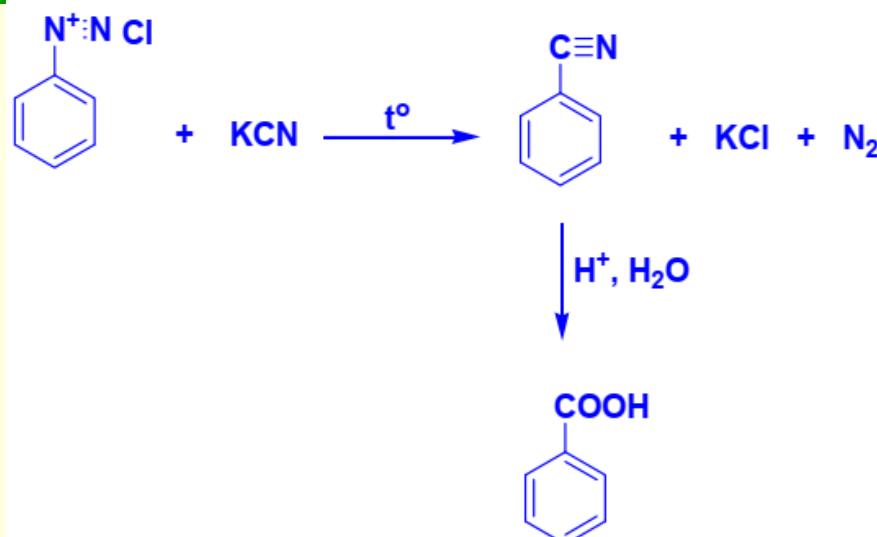


- Amine bậc 3
(chỉ có amine thơm)

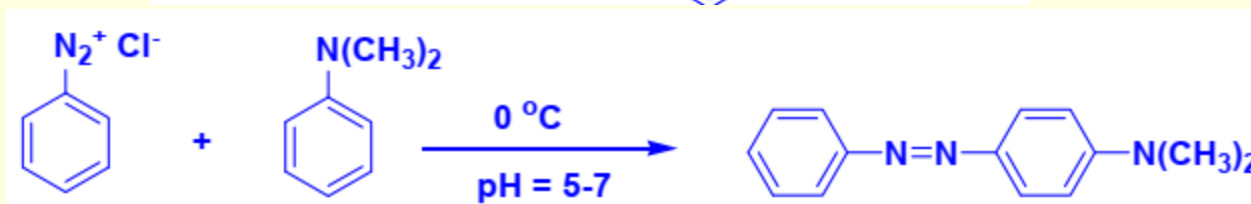




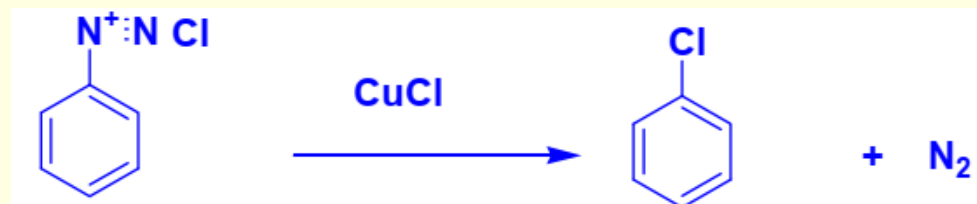
- Nitrile hóa



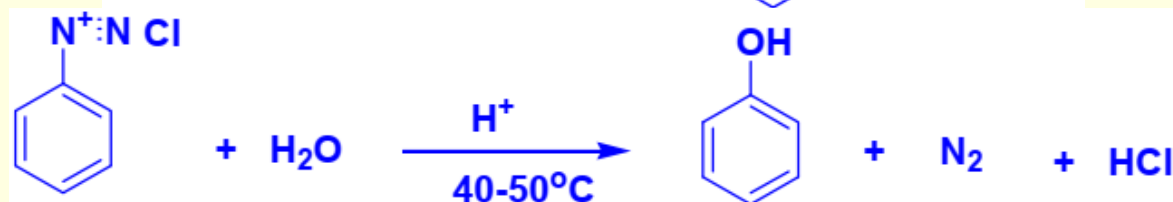
- Ghép đôi diazonium



- CuBr, CuCl



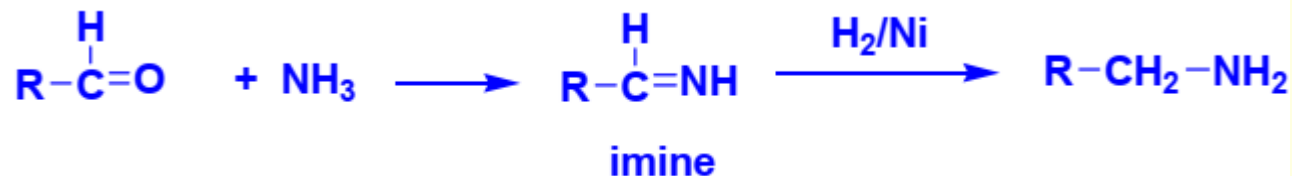
- +H2O



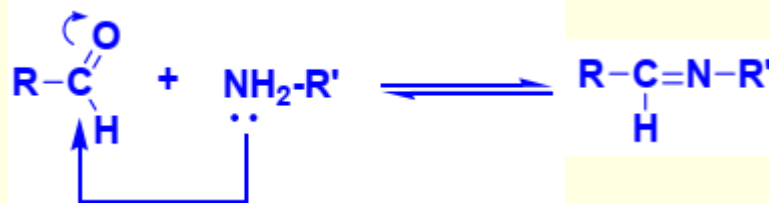
Phản ứng với aldehyde/ketone



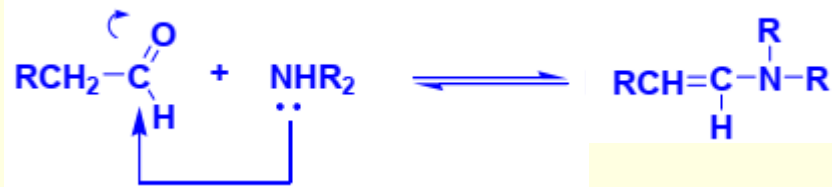
- NH_3



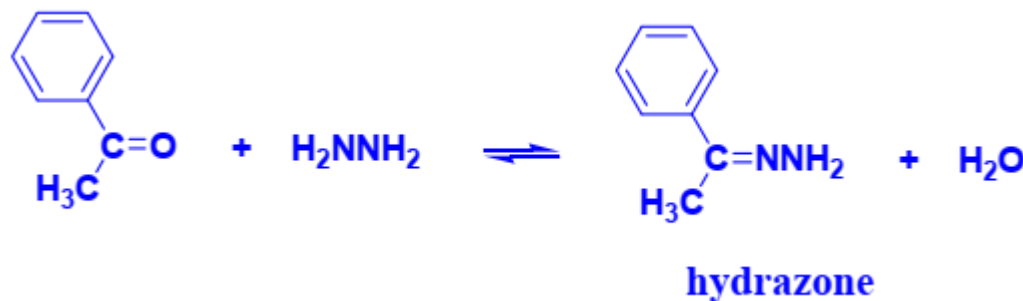
- Amine bậc 1



- Amine bậc 2



- Acetophenone



Summary



▪ Điều chế:

- Alkyl hóa Amine: amine + $RX \rightarrow -NHR, -NH_2R, -NR^+X^-$
- $NO_2 + H_2$ mới sinh (Fe/Sn + HCl) $\rightarrow NH_2$
- $R-CONH_2 + Br_2/OH^- \rightarrow R-NH_2$
- Tính base nên phản ứng với acid $RNH_2 + HX \rightarrow RNH_3^+X^-$
- Ketone/aldehyde + $NH_3/NH_2R \rightarrow -C=NH/-C=NR$
- $-CH_2-CH=O + -NHR_2 \rightarrow -CH=CH-NHR_2$
- Muối diazonium
 - $C_6H_5NH_2$ (aniline) + $NaNO_2 + HCl \rightarrow C_6H_5N^+\equiv N$
 - $C_6H_5N^+\equiv N + Y^-$ ($OH^- \rightarrow$ phenol, $CN^- \rightarrow C_6H_5CN$, thủy phân tạo C_6H_5COOH)
 - $C_6H_5N^+\equiv N + CuX \rightarrow C_6H_5X$
 - $C_6H_5N^+\equiv N + C_6H_5-(\text{nhóm đẩy điện tử}) \rightarrow$ vị trí p

Ví dụ

